BUSINESS REQUIREMENT DOCUMENT (BRD)

Project Title

KYC Smart Validator – Enterprise Customer Onboarding System

Project Type: Full Stack Web Application

Domain: Banking / Financial Services / Customer Management

Technologies Used: HTML5, CSS3, JavaScript, Bootstrap, Spring Boot (Java), MySQL, Maven, Postman

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Table of Contents

16. Conclusion

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1. Executive Summary

Overview

KYC Smart Validator – Enterprise Customer Onboarding System is a full-stack web application designed to automate the Know Your Customer (KYC) verification process during customer registration. The application validates customer PAN and Aadhaar information before registration to ensure only authentic and verified customer records are stored in the database.

The system helps organizations reduce manual verification efforts, eliminate duplicate customer records, improve data accuracy, and securely manage customer information through a centralized platform.

Importance of Digital KYC

Digital KYC has become an essential part of modern organizations because it enables faster customer onboarding, minimizes paperwork, improves regulatory compliance, reduces human errors, enhances customer experience, and provides secure management of customer data.

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2. Business Problem

Many organizations still rely on manual KYC verification, which creates several operational challenges.

• Manual verification is time-consuming and delays customer onboarding.

• Human errors during data entry lead to incorrect customer information.

• Duplicate customer records increase database inconsistency.

• Managing customer information manually becomes difficult as the customer base grows.

• Organizations require a secure and automated verification system to improve efficiency and maintain compliance.

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3. Proposed Solution

The KYC Smart Validator provides a complete digital solution for customer onboarding by validating customer identity before registration.

The system includes the following features:

Customer Registration

Allows administrators to register new customers through a secure web interface.

PAN Validation

Automatically validates the PAN number using predefined format validation rules.

Example:
ABCDE1234F

Aadhaar Validation

Validates the Aadhaar number using the Verhoeff Checksum Algorithm to ensure mathematical correctness before storing customer information.

Customer History

Displays previously registered customer records for easy search and management.

Live Dashboard

Provides real-time statistics including total customers, today's registrations, valid registrations, and rejected records.

REST API Integration

Uses Spring Boot REST APIs to enable communication between the frontend and backend.

Secure MySQL Database

Stores validated customer information securely in a MySQL database.

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4. Business Objectives

The primary objectives of the project are:

• Automate customer onboarding.

• Improve PAN and Aadhaar verification accuracy.

• Prevent duplicate customer registrations.

• Reduce manual verification effort.

• Improve operational efficiency.

• Maintain centralized customer records.

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5. Project Scope

In Scope

• Administrator Login

• Customer Registration

• PAN Validation

• Aadhaar Validation

• Dashboard

• Customer History

• MySQL Database

Out of Scope

• OCR Document Scanning

• Face Verification

• DigiLocker Integration

• SMS/Email OTP Verification

• Cloud Deployment

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6. Stakeholders

Administrator
Responsible for managing customer registration and monitoring system activities.

Customer Service Team
Registers customers and performs KYC verification.

Organization Management
Monitors business reports and customer onboarding performance.

System Administrator
Maintains the application, server, and database.

Customers
Provide PAN and Aadhaar information for verification and registration.

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7. Functional Requirements

FR-01: Administrator Login

The administrator should be able to securely log in to the system.

FR-02: Register New Customer

The system should allow administrators to register new customers.

FR-03: Validate PAN

The system should verify the PAN number format before registration.

FR-04: Validate Aadhaar

The system should validate Aadhaar using the Verhoeff checksum algorithm.

FR-05: Prevent Duplicate Records

The system should prevent registration if a PAN or Aadhaar number already exists.

FR-06: Store Customer Information

The system should securely store validated customer information in MySQL.

FR-07: Display Customer History

The administrator should be able to view all registered customers.

FR-08: Display Dashboard Statistics

The dashboard should display customer statistics and registration summaries.

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8. Non-Functional Requirements

User-Friendly Interface

The application should provide an intuitive and responsive interface.

Performance

The system should respond quickly during customer registration and validation.

Security

Customer information should be securely stored and protected against unauthorized access.

Reliability

The application should consistently perform customer validation without failures.

Scalability

The application should support increasing customer registrations with minimal performance degradation.

Maintainability

The application should follow modular coding standards to simplify future enhancements and maintenance.

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9. System Architecture

```
Frontend
(HTML, CSS, JavaScript, Bootstrap)
            |
            V
REST APIs
(Spring Boot)
            |
            V
Business Logic
(PAN & Aadhaar Validation)
            |
            V
MySQL Database
```

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10. Technology Stack

Frontend

• HTML5

• CSS3

• JavaScript

• Bootstrap

Backend

• Spring Boot (Java)

Database

• MySQL

Development Tools

• Maven

• Postman

• Visual Studio Code

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11. Project Workflow

```
Administrator Login
      |
      V
Dashboard
      |
      V
Customer Registration
      |
      V
```

PAN Validation
|
V
Aadhaar Validation
|
V
Store Customer Data
|
V
Customer History

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12. REST APIs

POST /api/customers/register

Registers a new customer.

POST /api/validate/pan

Validates the PAN number.

POST /api/validate/aadhaar

Validates the Aadhaar number using the Verhoeff checksum algorithm.

GET /api/customers/history

Retrieves customer registration history.

GET /api/dashboard/stats

Returns dashboard statistics.

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13. Database Design

Customer Table

Customer_ID (Primary Key)

Name

Email

Mobile_Number

PAN_Number

Aadhaar_Number

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14. Business Benefits

• Faster customer onboarding process.

• Reduced manual verification effort.

• Prevention of duplicate customer records.

• Improved PAN and Aadhaar validation accuracy.

• Secure customer information storage.

• Better customer experience through faster registration.

• Easy management of centralized customer records.

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15. Future Enhancements

The project can be extended with the following features:

• OCR-Based PAN and Aadhaar Scanning

• Face Verification

• DigiLocker Integration

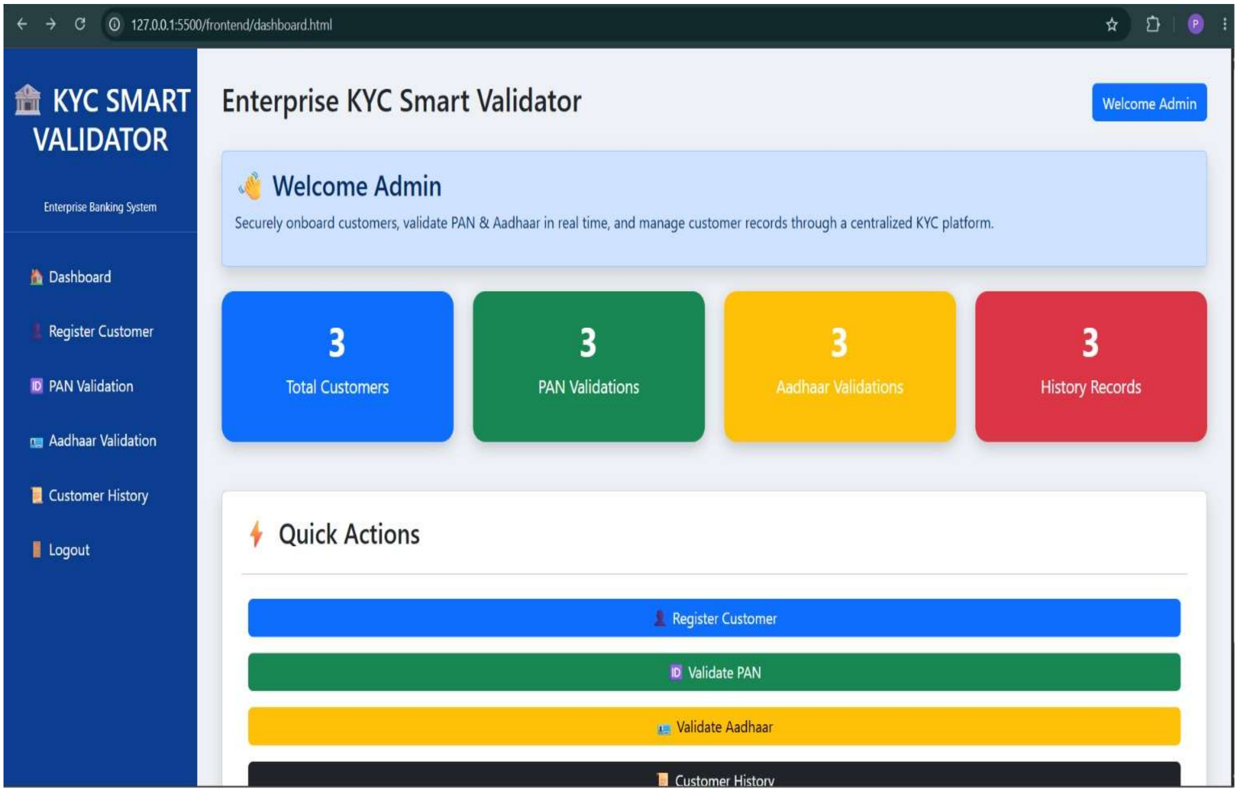
• AI-Based Fraud Detection

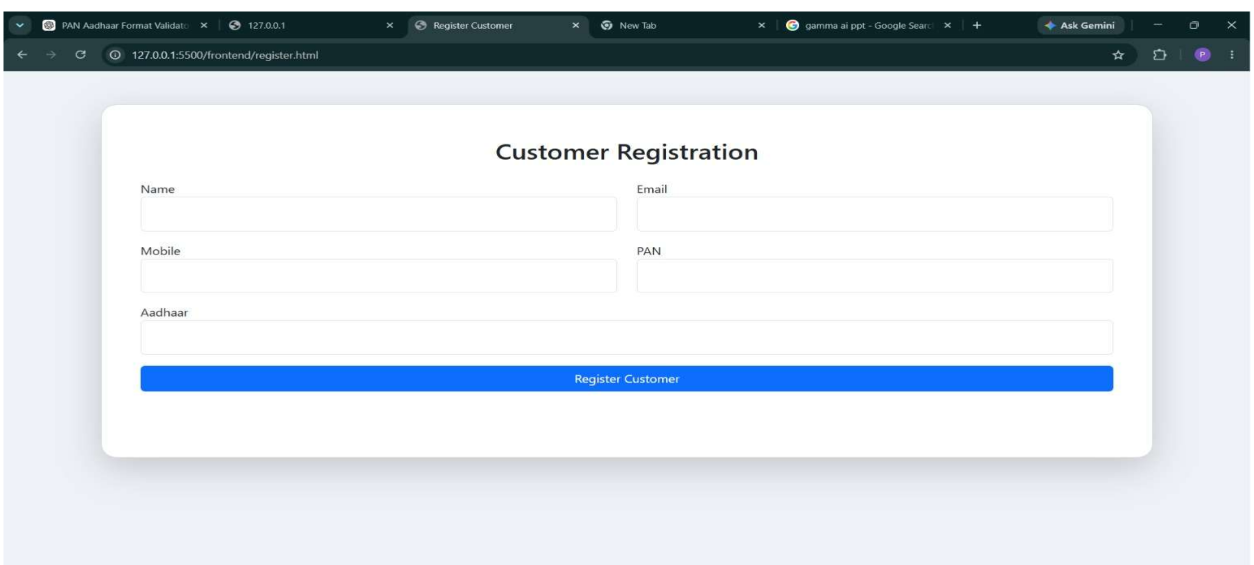
• Email and SMS OTP Verification

• Role-Based Authentication

• Advanced Analytics Dashboard

Output:





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16. Conclusion

KYC Smart Validator – Enterprise Customer Onboarding System is an enterprise-ready web application that automates the customer onboarding process by validating PAN and Aadhaar information before registration. The system significantly reduces manual effort, minimizes human errors, prevents duplicate customer records, and securely stores customer information in a centralized MySQL database.

Developed using HTML, CSS, JavaScript, Bootstrap, Spring Boot, MySQL, Maven, and Postman, the project follows a modern RESTful architecture that ensures scalability, reliability, and maintainability. With future enhancements such as OCR, AI-based fraud detection, DigiLocker integration, OTP verification, and role-based authentication, the application has the potential to evolve into a comprehensive enterprise KYC solution.

Project Summary

Project Name: KYC Smart Validator – Enterprise Customer Onboarding System

Project Type: Full Stack Web Application

Frontend: HTML5, CSS3, JavaScript, Bootstrap

Backend: Spring Boot (Java)

Database: MySQL

Build Tool: Maven

API Testing: Postman

IDE: Visual Studio Code

Architecture: RESTful Multi-Layer Architecture

Primary Goal: Automate customer onboarding through PAN and Aadhaar validation.

Target Users: Administrators, Customer Service Teams, Organizations

Expected Outcome: Faster onboarding, improved verification accuracy, reduced manual effort, secure data storage, and efficient customer management.